

Broadband multimedia delivery over copper

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Asymmetric digital subscriber loop (ADSL) technology has proved itself in various telco video on demand (VoD) trials as a technically viable means of delivering VHS quality moving video over the existing twisted copper pairs of the access network. A very high-speed version of this technology (VDSL) is now emerging which has the potential to deliver more channels of higher quality broadband multimedia services over a hybrid fibre-copper access network. This paper provides an overview of the enabling copper transmission technologies (both ADSL and VDSL) and the metallic access environment.

1. Introduction

Telecommunications operators' fixed networks were originally constructed entirely from metallic transmission media (predominantly copper), carrying voice-band signals in the region 300 Hz to 3.4 kHz. In developed countries, the core or backbone of the network that interconnects the switching centres is now mainly optical fibre. However, the access portion of the network that connects switches to customers (the so-called 'last mile') is still dominated by twisted copper pairs. There are over 560 million twisted copper pair connections world-wide. The UK alone has around 27 million copper access connections with a replacement cost of over £11 billion. Optical access network technologies have been developed and proven; however, the sheer inertia of the installed copper base means that it could take many years to migrate the access networks from copper to fibre. This paper describes the transmission technologies available to telcos to deliver broadband services such as TV or video on demand (VoD) over the existing copper access network as it evolves towards a fibre future.

2. The copper access network transmission environment

This section overviews the metallic access network. In particular, it describes recent developments in transmission channel modelling. It also examines impairments and the theoretical information carrying capacity of the metallic access network.

2.1 Access network topology

Figure 1 shows an overview of BT's access network. The main network consists of large multi-pair cables which radiate out from the main distribution frame (MDF) located within the serving local exchange to flexibility points

known as primary crossconnection points (PCPs). Individual cable segments are joined together to form the link from the MDF to PCP. From the PCP outward, the network is known as the distribution network (D-side) whereas the MDF to PCP link is known as the exchange (E-side) network. From the PCP connections radiate out to distribution points (DPs) sometimes passing through secondary crossconnection points (SCPs) for increased flexibility. From the DP the connection is made to the customer's premises via the dropwire, or final drop. It should be noted that the distribution network may be overhead or underground. Underground feed is more reliable and is usually used in modern urban areas. However, in rural areas the high civil engineering costs associated with underground cabling prohibit the use of underground feeds and overhead feeds are more common. Overhead feeds are also used in older conurbations.

At the customer's premises there is a demarcation point known as the network termination point (NTP). This is the point at which BT's network officially ends and is a point at which service is provided to the customer via a standardised user/network interface (UNI). For a customer receiving telephony service (POTS) this point is the familiar white master socket.

The cables that are predominantly used in the access network are unshielded twisted pairs (UTPs) of differing gauges. The smaller gauges are found closer to the exchange (to make the large multi-pair cables easier to handle) and the larger gauges are found toward the customer's premises where the environment is more hostile and there is a need to achieve maximum range for a given transmission and signalling resistance requirement. Copper cables dominate, but aluminium cables are also found in BT's network. Typical gauges of cables (from MDF to DP) are 0.32, 0.4, 0.5, 0.6, 0.63, 0.7, 0.8, and 0.9 mm.

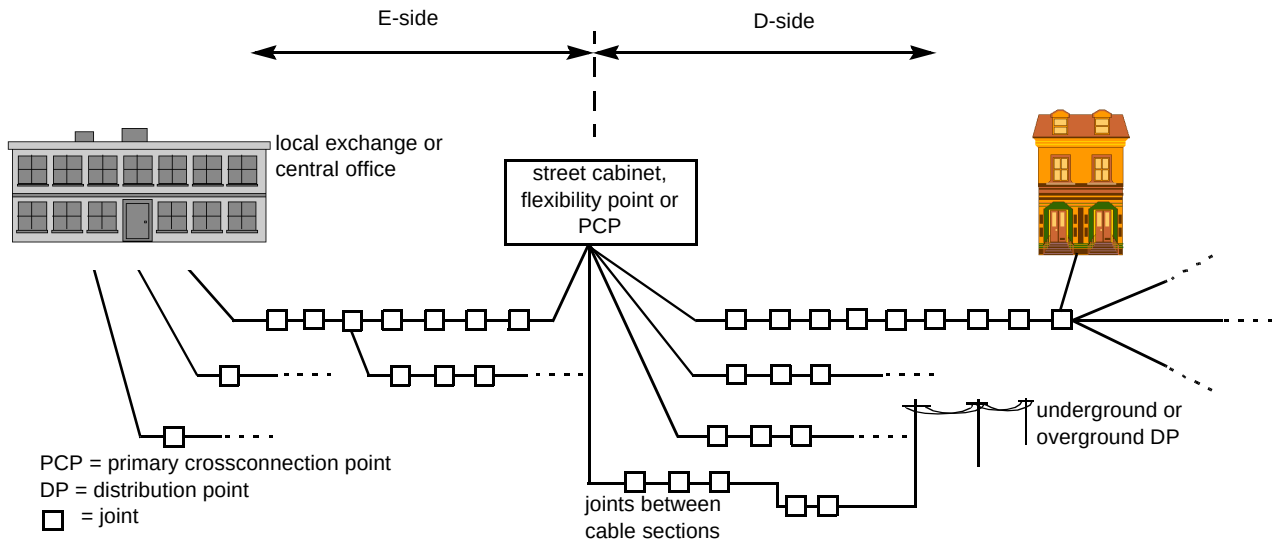


Fig 1 Access network topology.

2.2 Advances in transmission channel modelling

The metallic twisted wire pair was originally specified and designed to carry analogue telephony signals within a 300 Hz to 3.4 kHz frequency band. As frequency rises so does the signal attenuation due primarily to the 'skin effect' which manifests itself as an increasing resistive component per unit length. Most of BT's access cables are made from solid copper conductors sheathed with polyethylene (PE) dielectric. This forms a low but significant capacitance per unit length. However, for most polyethylene cables capacitance per unit length is constant and does not vary appreciably with frequency.

It is useful to be able to model the electrical behaviour of the access channel when designing and estimating performance of various transmission technologies. Recent work at BT Laboratories [1, 2] has led to the development of an empirical model for homogeneous access channels. The model has two forms — firstly, a 7-parameter form covering the region from DC to 2.5 MHz (i.e. ADSL operation), secondly, an enhanced 13-parameter version covering the frequency range DC to 30 MHz (for high-rate studies such as VDSL).

The 7-parameter empirical model

Figure 2 shows a homogeneous transmission line in a circuit of equal source and load impedance Z_c (in equivalent unbalanced form).

The characteristic impedance (Z_c) and propagation constant (γ) are known as the secondary line constants and

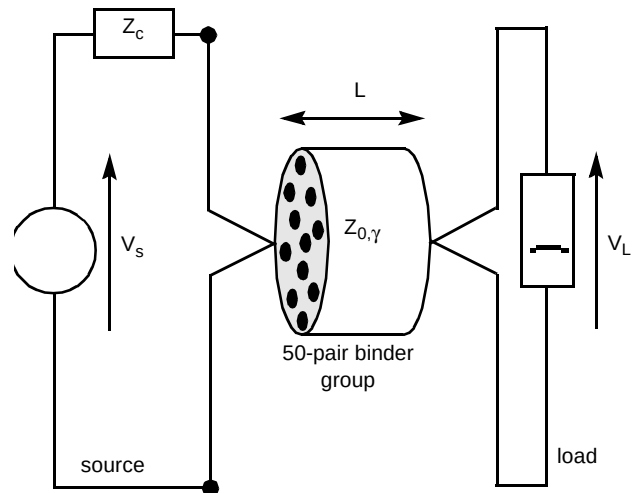


Fig 2 Twisted pair channel.

may be defined in terms of the primary line constants R , L , C and G :

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$\gamma = \sqrt{(R + j\omega L) \cdot (G + j\omega C)}$$

R is the resistance per unit length (usually Ω/km), C is the capacitance per unit length (usually nF/km), L is the inductance per unit length (usually $\mu\text{H}/\text{km}$) and G is the conductance per unit length (usually $\mu\text{S}/\text{km}$).

Figure 3 shows the lumped parameter representation of a small length of a metallic access channel. The primary line

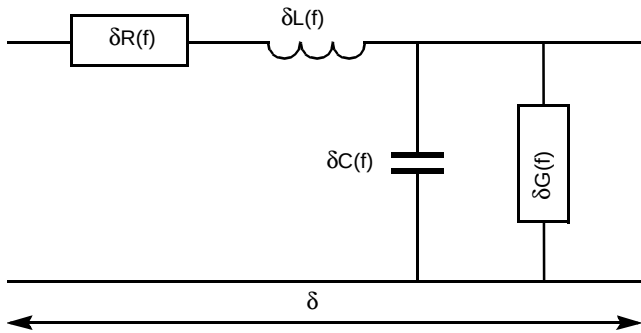


Fig 3 Lumped parameter representation.

constants R and L are shown in unbalanced form. As $\delta \rightarrow 0$ an analytic expression for the channel voltage transfer characteristic may be found in terms of the secondary (and hence primary) line constants. Hence the complex amplitude and phase response of the channel may be calculated from knowledge of the behaviour of the primary line constants which characterise the channel.

Recent research at BT Laboratories has shown that a relatively simple 7-parameter empirical channel model may be used to accurately predict the salient transmission characteristics of both D-side and E-side metallic access channels. The model has been verified against various lengths and gauges of both aluminium and copper pairs which are typical of BT's access network. Each metallic cable type has 7 characterising parameters. These parameters determine the variation of the primary line constants with frequency. The empirical model found for the primary line constants is described below:

$$R(f) \sqrt[4]{r_0^4 + a \cdot f^2} \quad (\Omega/\text{km})$$

where r_0 is the DC resistance and a is a constant which controls the onset of the $\sqrt{f}$ part of the curve.

$$L(f) = \frac{l_0 + l_m \left[\frac{f}{f_m} \right]^b}{1 + \left[\frac{f}{f_m} \right]^b}$$

where l_0 and l_∞ are the low- and high-frequency inductance respectively, and f_m and b are parameters which control the transition between the two regions.

$$C(f) = k \quad (\text{F/km})$$

where k is a constant.

$$G(f) = 0 \quad (\text{S/km})$$

the ultimate simple model.

This model has enabled computer simulations of arbitrary channel response to be made quickly and efficiently. This reduces the time needed to predict the

transmission performance of various transmission technologies. The model has also been used to determine the theoretical information-carrying capacity of arbitrary access channels.

The 13-parameter empirical model

For the study of very high-rate transmission over metallic access pairs, an enhancement to the 7-parameter model is made. This enhancement extends the frequency of operation of the model from 2.5 MHz to beyond 30 MHz. This enables both D-side and dropwire channels to be covered. The enhanced model is described below. The resistance per unit length (R) is given by:

$$R(f) = \frac{1}{\left[\frac{1}{\sqrt[4]{r_s^4 + ar \cdot f^2}} \right] + \left[\frac{1}{\sqrt[4]{r_{0s}^4 + br \cdot f^2}} \right]} \quad (\Omega/\text{km})$$

where r_0 is the DC resistance (in Ω) of the copper conductor, ar is a constant which controls the onset of the $\sqrt{f}$ part of the curve, r_{0s} is the DC resistance of the steel part of the inner conductor (for dropwire types) and br is a constant which again controls the onset of the $\sqrt{f}$ part of the curve. Note f is frequency (in Hz).

This model for R is different from the 7-parameter model described earlier. There are two resistive components which add in parallel to cater for the copper-plated steel conductors found in some dropwire types. Firstly, there is a component due to the copper outer conductor and a new component due to the steel inner core. For solid copper dropwires this expression reduces to the previous model.

The inductance (L) per unit length is given by the same equation as for the 7-parameter model, no elaboration having been found necessary.

The capacitance (C) per unit length is given by:

$$C(f) = c_\infty + c_0 \cdot f^{cb} \quad (\text{F/km})$$

where c_0 and c_∞ the low- and high-frequency capacitance respectively, and cb is a parameter which controls the transition between the two regions (note cb is negative). This enhancement allows the model to be used for the PVC dielectrics found in some dropwires.

The conductance (G) per unit length is given by:

$$G(f) = gb \cdot f^{gc} \quad (\text{S/km})$$

where the parameters g_b and g_c are constants. The 13-parameter model has been validated from DC to 30 MHz with excellent accuracy (typically 0.1 — 0.5 %).

Example dropwire channels

The insertion loss of a channel may be readily calculated from the empirical channel model according to the following relationship:

$$I(f) = 20 \cdot \log \left[\frac{2 \cdot Z_c \cdot Z_0}{Z_0^2 \sinh(\gamma l) + 2 \cdot Z_c \cdot Z_0 \cosh(\gamma l) + Z_c^2 \sinh(\gamma l)} \right]$$

Figure 4 shows a plot of the typical insertion loss of 300 m sections of various dropwire types used by BT in its access network. It can be seen that there is a large variation in insertion loss at the higher frequencies (e.g. >30 dB at 20 MHz). This is not surprising given the variation in physical construction. A very high-rate transmission system operating over these channels will have very different performance for a given channel length.

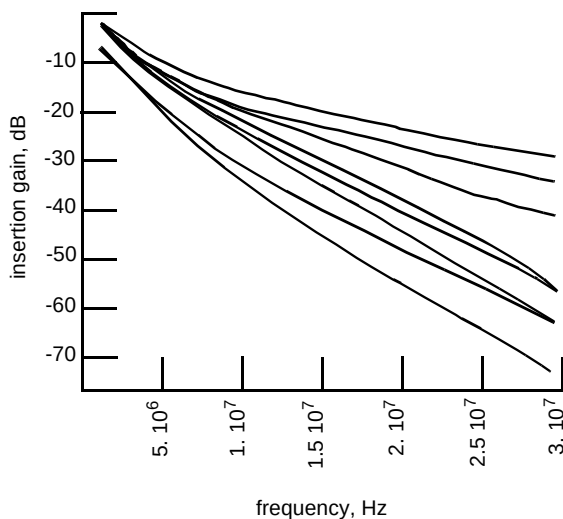


Fig 4 Insertion loss of 300 m lengths of various dropwire types.

2.3 Impairments

Without noise (or impairments) even a bandlimited access channel has infinite information-carrying capacity. In reality there are many sources of noise which collectively will determine the information-carrying capacity of the access channel. In a well-designed digital transmission system, the effects of internal system noise will be small. In this case the information-carrying capacity will be determined by the access channel and external noise sources, namely:

- near-end crosstalk (NEXT),

- far-end crosstalk (FEXT),
- radio frequency interference (RFI),
- impulsive noise.

Each of the above noise sources will, to a greater or lesser extent (in combination), determine the information-carrying capacity of the access channel. Systems which are required to deliver very low bit error rates (BERs) must be designed to operate in a noise environment dominated by these impairments (as well as the channel attenuation itself) [2].

Internal noise sources should not be ignored, but they will largely be:

- a feature of the state of developments in signal processing technology,
- an ever-present feature of the passive and active electrical components (e.g. white noise).

White noise

Generally, white noise sources are relatively low level and are usually insignificant when compared with other noise sources (e.g. NEXT and impulse noise). However, they are fundamental and rarely avoided because they are always present in the background. Thermal noise for example is generated by electronic amplifiers, analogue-to-digital converters and even passive line termination components (e.g. resistors). White noise may have thermal, shot and quantisation components.

Near-end crosstalk (NEXT)

NEXT arises due to signals from transmitters on other pairs operating in the same multi-pair cable. These are signals which interfere with the input of a transceiver at the same end. The transmitted signal leaks into the receiver via capacitive and inductive coupling paths (see Fig 5). NEXT is usually the most significant noise source which limits the reach of high-rate duplex transmission systems.

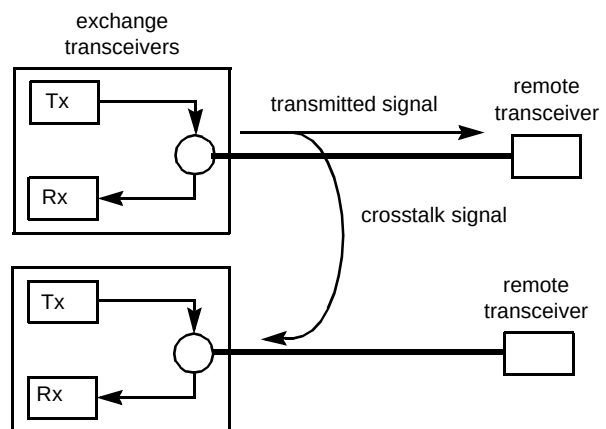


Fig 5 NEXT noise coupling.

Far-end crosstalk (FEXT)

FEXT occurs when signals from transmitters on other pairs in the same cable leak into the input of the wrong transceiver at the other end (see Fig 6).

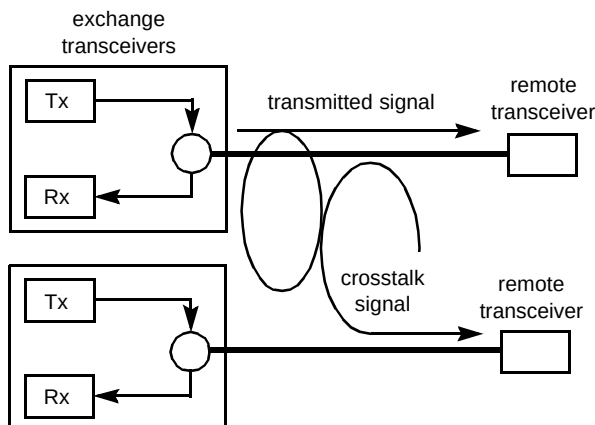


Fig 6 FEXT noise coupling.

FEXT is not usually significant for symmetrical digital transmission systems because the FEXT always suffers more attenuation than the signal passing through the channel. However, it is significant for ADSL systems where the different directions of transmission occupy different frequency bands. As for NEXT, FEXT is usually worst when the interfering signals on the other pairs are of the same kind as the system being interfered with. This is known as self-FEXT.

Radio frequency interference (RFI)

RFI is a source of noise which affects almost every access wire-pair. It varies considerably from pair to pair and has characteristics which show temporal variations. Digital

transmission systems must tolerate certain levels of RFI as prescribed by current European legislation on electromagnetic compatibility (EMC).

Varying amounts of RFI will undoubtedly exist in the same frequency range as the transmitted signal. This interference is in-band and therefore cannot be filtered out. Metallic access transmission systems should not interfere with radio transmissions. This places a limit on the transmitted power spectral density. It is fortunate that access network pairs are well balanced and do not pick up or radiate RFI easily. The longitudinal balance (common mode to differential coupling loss) of most wire-pairs is extremely large at low frequencies (> 60—70 dB) decreasing to around 30 dB at 30 MHz. This means that RF signals will suffer quite large attenuation before coupling into the receiver of a transmission system.

RFI is an important noise source for very high-rate transmission systems which occupy the same bands as amateur radio enthusiasts (1.8 to 30 MHz). This type of RFI is difficult to cancel adaptively because of the on/off nature of single sideband suppressed carrier transmission.

Figure 7 shows typical levels of RFI induced in the band DC to 1 MHz on an underground wire-pair delivering ordinary telephony.

Impulse noise

Impulsive noise is another key impairment for advanced access transmission on metallic twisted pairs. It is caused by a variety of sources producing short electrical transients. Such transients may come from:

- household appliances switching on and off (e.g. fluorescent lights, refrigerators, cookers, etc),

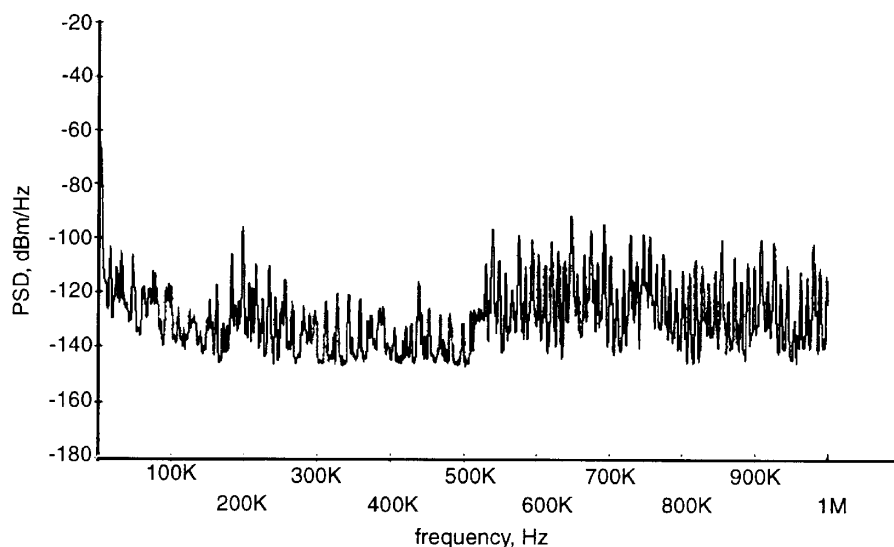


Fig 7 RFI induced in a terminated underground wire-pair.

- telephony itself (e.g. on-hook/off-hook and ringing),
- arc welders, electric cattle fences or locomotive traction systems, etc.

These disturbances can be electromagnetically coupled into the access network and may cause error bursts in digital transmission.

Forward error correction (FEC) and data interleaving are normally used in ADSL and VDSL systems to mitigate the effects of impulse noise. However, there is a trade-off — increasing protection against impulse noise is usually accompanied by increased transmission delay or latency.

2.4 Theoretical capacity

The theoretical information-carrying capacity (in the presence of Gaussian noise) of an access channel may be calculated using the well-known Hartley-Shannon theorem:

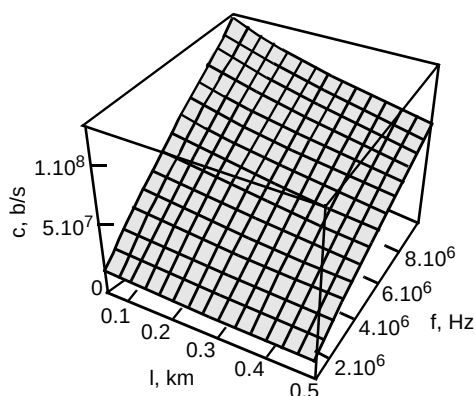
$$C = B \cdot \log_2 \left(1 + \frac{S}{N} \right) \text{ bit/s}$$

where B is the bandwidth, and S/N is the mean-square signal-to-noise ratio (SNR) at the receiving station in the channel.

Assuming a base-band signal, then the information-carrying capacity is given by:

$$C = \int_0^{f_{\max}} \log_2 \left(1 + \frac{S}{N} \right) \text{ bit/s}$$

Using the empirical channel model and knowledge of the behaviour of the more predictable noise sources (e.g. crosstalk), the theoretical maximum information capacity may be calculated for arbitrary channels and noise environments. Extensive simulation work has been carried out at BT Laboratories.



With respect to Gaussian noise, both ADSL and VDSL systems are primarily FEXT-limited because they may employ FDM techniques to separate the two directions of transmission (thus removing the unwanted effects of NEXT) — 1% worst-case FEXT is easy to predict given knowledge of the power spectral densities of the interferers and the channel length.

A study of the information capacity of various metallic cable types is quite revealing [2]. For example, Fig 8 shows the FEXT-limited capacity of 0.5 mm D-side underground cabling when appropriate assumptions are made for a VDSL transmission environment. Capacity is plotted as a function of the maximum upper frequency (of a baseband transmission) and channel length. Contours of fixed capacities of 100, 50 and 25 Mbit/s are also shown. The potential for very high-rate transmission over relatively short distances is clearly seen. It must be noted that this theoretical limit will only be approached with transmission systems of infinite complexity and presumably infinite cost. Practical ADSL systems are achieving typically 30—60% of theoretical capacity predicted by knowledge of the channel and noise environment. VDSL systems are expected to realise similar levels of practical capacity. Practical systems will also be limited by the effects of other (less tangible) noise sources such as RFI and impulse noise. However, there is obvious potential for exploitation of the existing metallic access network by systems such as ADSL and VDSL.

3. Overview of transmission systems

3.1 The ADSL concept

ADSL is a method of transmitting digital data at high bit rates over the existing installed twisted pairs of the access network. This technology will release the previously untapped capacity in the copper telephone network as

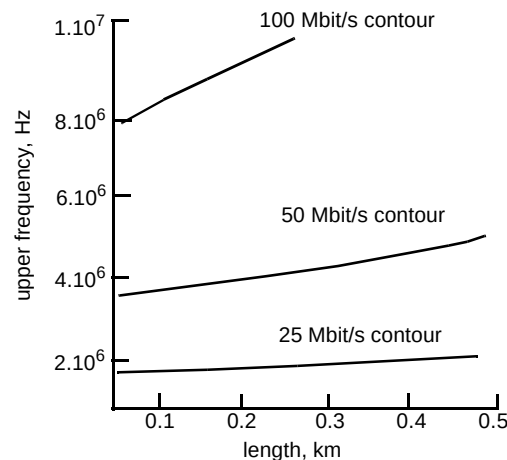


Fig 8 Theoretical capacity of 0.5 mm D-side cabling.

described in the previous sections. When the transmission capability of ADSL is combined with the latest generation of video compression technology, delivery of broadband interactive multimedia services over the copper access network becomes possible.

Voiceband modems are now able to transport 28 kbit/s through a 4 kHz voice channel. Use of similar advanced modulation and signal processing techniques have given ADSL technology the power to deliver downstream (exchange-to-customer) transmission channels of up to 6.144 Mbit/s. This is achieved without disturbing the telephony service already installed on the line. A low-speed return channel (customer to exchange) enables customers to select and control the downstream information. An ADSL transceiver can be installed simply at each end of a customer's existing copper line (Fig 9). Power is then applied and within seconds the customer has a digital link to the telco's network. New interactive services can then be delivered to customers over their existing phone lines.

Marketable interactive residential broadband services are still at an embryonic stage, yet it is already clear that many will be asymmetric in terms of data rate requirement, with a high capacity needed towards the customer and a low capacity in the return direction. Video-on-Demand is attracting much interest because many customers can identify with the concept of having the equivalent of a video rental store combined with a VCR in the network. It would allow customers to select, stop, start, pause, rewind and fast-forward video films. An ADSL access transmission system, operating at 2.048 Mbit/s, can deliver MPEG (Moving Picture Experts Group) algorithm digitally encoded video, plus digital stereo sound. Such a system has a subjective picture quality comparable to that of a domestic VCR. A system operating at 6.144 Mbit/s can deliver three VHS quality channels or be dedicated to a single MPEG encoded session with a subjective picture quality comparable to live broadcast. Other interactive visual services such as home shopping, games and education might use the same network infrastructure.

The ADSL concept and its application to transport interactive video services was proposed at the beginning of

the decade by research workers such as Lechleider [3] and Cioffi [4]. It has since evolved from computer simulations and laboratory prototypes to production systems which are available today and will become increasingly integrated. The principle is simple — transmit a high-capacity channel downstream to the customer and transmit at a much lower bit rate in the return direction from the customer to the network, and then do this simultaneously and without interfering with existing telephony on the copper pair.

Since the return transmitter operates at a much lower frequency compared with the forward direction (see Fig 10), the levels of crosstalk at the customer end are low compared to symmetrical data rate systems [5]. The absence of such crosstalk enables a much greater transmission distance to be achieved.

By employing a passband modulation method an ADSL transceiver can operate at frequencies above those of standard telephony and, provided filtering is employed to remove undesirable noise transients such as those due to loop disconnect and ringing, then an ADSL system can co-exist with telephony on the same pair. Intelligent signal processing in the ADSL transceiver can adapt automatically to optimise performance on each individual copper line and track temporal changes, such as those due to temperature, moisture or continuous interference sources. Manual setting up is also avoided. Provided the line is within the permissible planning limit, the transceivers start up automatically by means of a training sequence which lasts a few seconds.

There exists a family of standardised ADSL systems, with downstream bit rates from 1.536 to 6.144 Mbit/s. The bit rates for the simplex downstream channels are based on multiple channels with a maximum total capacity of 6.144 Mbit/s. This allows up to four possible transport channels of 1.536 Mbit/s or up to three channels of 2.048 Mbit/s. The ADSL data multiplexing format of the American National Standards Institute (ANSI) ADSL standard [6] has been designed to be flexible enough to allow other channelisations, for example to transport data in ATM format. Some manufacturers' ADSL implementations will deliver around 8 Mbit/s downstream to accom-

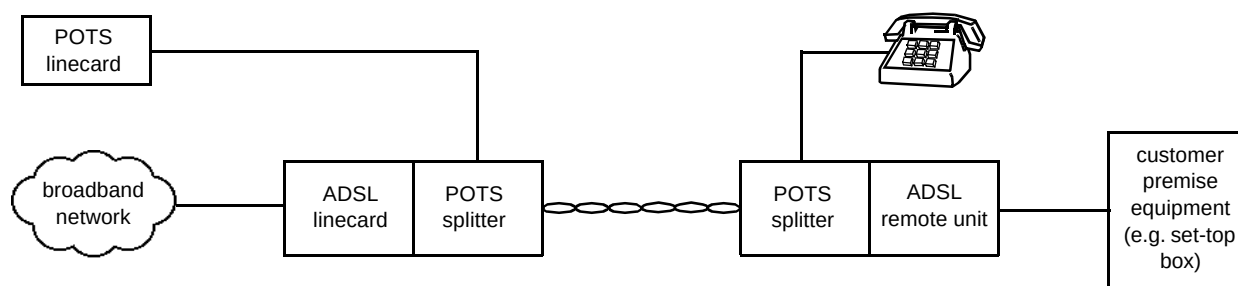


Fig 9 ADSL in the access network.

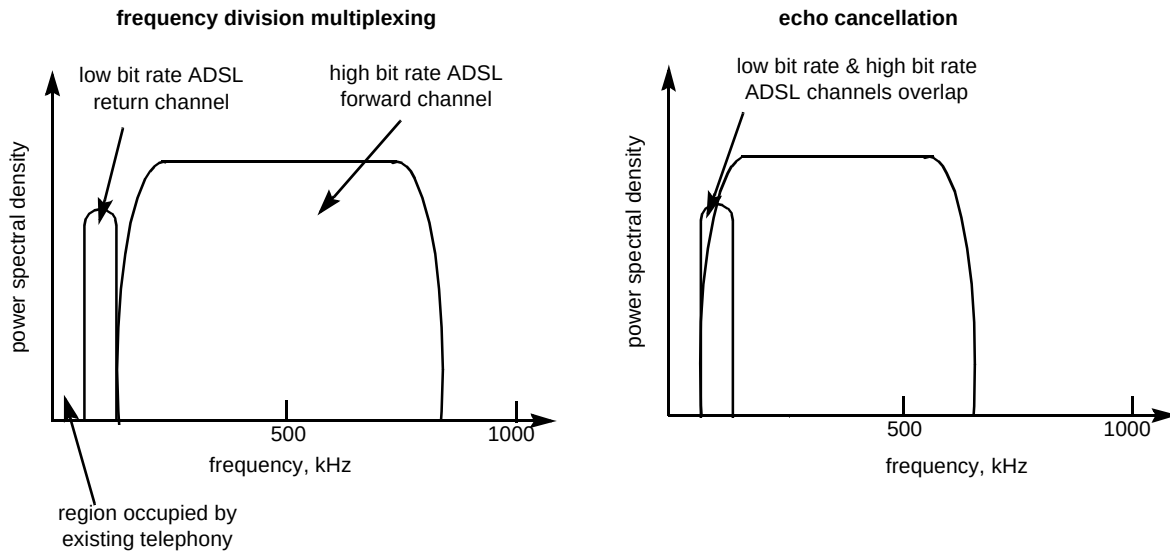


Fig 10 ADSL channel separation.

modulate four channels of 2.048 Mbit/s. The transmission reach of a system decreases as the operating bit rate is increased. ADSL systems operating at 2.048 Mbit/s are expected to have around 90% of BT customers within range. However, systems operating at the higher rates have a reduced range and thus a lower network penetration.

In addition to the simplex downstream channel, an ADSL system will transport a return or control (C) channel for user-to-network signalling. The standard C channel operates at either 16 kbit/s or 64 kbit/s, although some early systems use 9.6 kbit/s. An embedded operations channel is used to carry network management information.

Two further optional, narrowband, duplex channels may be transported simultaneously by an ADSL. These can have additional capacities of 160 kbit/s and 384 kbit/s. Maximum time delay limits on some services place restrictions on the degree of latency allowed in the ADSL system. For such services, delay is minimised by dispensing with the interleaving of the error correction code.

3.2 The hybrid fibre/copper concept

In time the telco copper networks will begin to include more optical fibre cables, thus ensuring that homes have the best possible connectivity for future communications needs. This will be required in the future as interactive multimedia services proliferate. Each member of a family may wish to access such services simultaneously, e.g. one member watching a movie, another shopping from home and children playing games or getting homework assistance over the network.

Researchers are now studying even higher speed access systems for transmission over the shorter D-side copper wires at the customer end of the access network. Work on these high-speed systems is focused on two scenarios for a hybrid fibre/copper access network [7], namely fibre-to-the-cabinet and fibre-to-the-kerb. This technology is being tagged very high rate ADSL (VADSL). This can use the same transmission techniques developed for ADSL. However, due to the shorter lengths of copper involved in only connecting from the customer to a kerbside cabinet, higher transmission rates are achievable. In the fibre-to-the-kerb architecture, optical fibre is deployed as far as the DP or last joint box in the D-side of the network (refer to Fig 1). This requires the VDSL transmission system to operate over 'copper tail' distances of around 50 m to 500 m (Fig 11). Manufacturers and standards fora are examining data rates in the range 25 Mbit/s to 51 Mbit/s for this scenario. The second architecture being considered for VDSL is a hybrid fibre/copper network where optical fibre is only deployed as far as the cabinet. This requires VDSL transmission over a distance of around 1 km. Data rates up to 25 Mbit/s are being examined for this fibre-to-the-cabinet architecture. Downstream data rates of around 12, 25 and 51 Mbit/s are of interest because of their relationship to transport rates being examined by the ATM Forum. Upstream data rates are mainly focused on the range 500 kbit/s to around 2 Mbit/s. Some initial investigations to compare the two hybrid fibre/copper architectures have shown fibre-to-the-cabinet as being the more cost effective [8].

Remotely sited VDSL using an optical fibre feed, could remain an access network option for some years. This high-speed copper transmission could play a significant role in bringing multiple channels and higher quality channels, such as those required for HDTV, to those customers who require them.

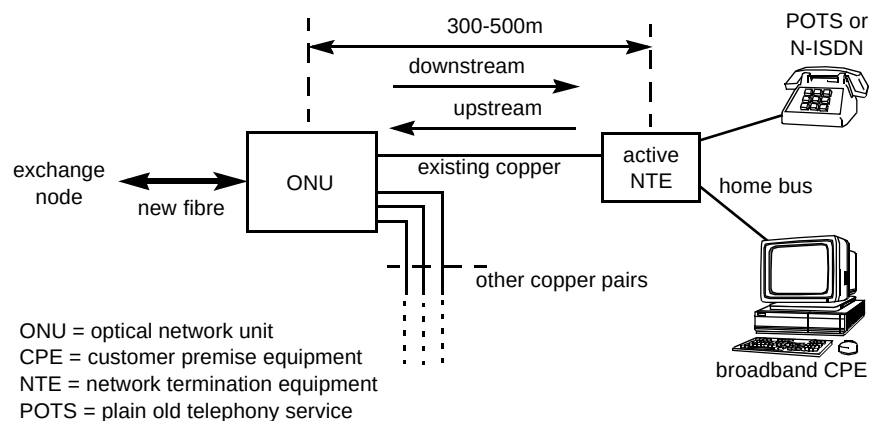


Fig 11 Hybrid fibre/copper access network.

4. Transmission techniques

4.1 Separation of downstream and return channels

Two directions of transmission have to be achieved on the same twisted pair. Early ADSL systems exploit frequency division multiplexing (FDM) as shown in Fig 10. The low bit rate control channel usually occupies the frequency range immediately above the voiceband region of telephony. The high bit rate forward or downstream channel occupies the higher frequencies. The total operating frequency range for ADSL is usually limited to about 1 MHz by encoding several bits per symbol. The channels in Fig 10 are shown with flat spectral distribution. In reality this may be varied in order to maximise performance. For VDSL the spectra could theoretically extend as high as 30 MHz, constrained by EMC considerations. Practical systems will, however, probably not extend much beyond 10 MHz. The

VDSL spectra may start at a much higher frequency than ADSL, e.g. nearer to 1 MHz instead of 20 kHz. There are three advantages to this:

- it could avoid some radio bands used for AM broadcast transmissions, thus avoiding a potential RFI noise source in certain locations,
- it allows the use of simpler (and hence smaller) filters to split the VDSL signals from baseband services such as POTS — this is important since one end of the system would be located in an external street unit,
- it allows both POTS and basic rate ISDN systems to exist below the VDSL signals.

This facilitates graceful network evolution by initially using the hybrid fibre/copper network as a broadband overlay (Fig 12). Then existing services, such as basic rate

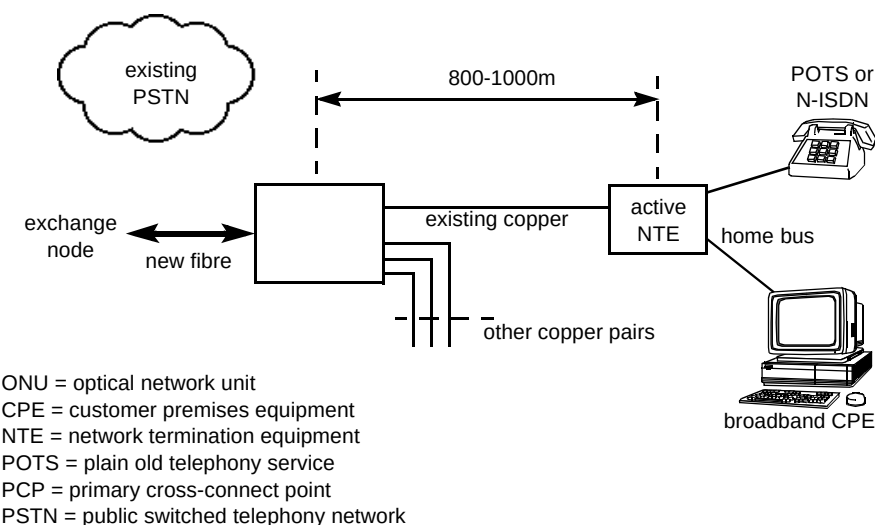


Fig 12 Hybrid fibre/copper broadband overlay.

ISDN, could later be migrated by embedding them within the payload of the hybrid fibre/copper broadband delivery system.

For ADSL there is current interest in allowing the high bit rate forward channel to overlap with the low-speed return channel (Fig 10), in order to take advantage of the reduction in cable loss at the lower frequencies. This may be achieved using asymmetric echo cancellation which models and cancels the interference between forward and return channels. The ADSL standard [6] allows for interworking between different systems by establishing either the FDM or echo-cancellation method during the start-up sequence. Asymmetric echo cancellation is particularly new for discrete multi-tone (DMT) ADSL transceivers and many manufacturers are choosing to omit this option in their initial developments. The additional complexity of echo cancellation is expected to yield around 2 dB of performance improvement. The use of echo cancellation effectively moves part of the filtering problem from analogue to digital signal processing which is more amenable to VLSI. The benefits of echo-cancelling ADSL systems are greater when the control-channel bandwidth is expanded to carry a larger payload such as 384 kbit/s videotelephony. In such circumstances an FDM implementation forces the downstream channel to occupy higher frequencies with subsequent increased attenuation and hence reduction in range. It should be noted that overlapping the two ADSL channels by the use of echo cancellation introduces the possibility of an additional noise source known as self-NEXT (see section 3.3), which does not occur in FDM implementations. This will be more acute for large-scale ADSL deployment where the possibility of other ADSL crosstalk interferers is increased. However, as long as the transmit PSD is controlled to be at a similar level

to other systems occupying the overlap frequencies (e.g. HDSL) and the return channel bandwidth is not in excess of these other systems, then the worst-case multiple-crosstalk models used in many specifications, standards and deployment planning rules will still be valid.

The coexistence of ADSL on the same line as existing telephony presented a major technical challenge. ADSL signals of less than 1 mV need to be recovered in the presence of ringing and loop disconnect pulses of many tens of volts. Hence the design of the separation filter requires very high stopband attenuation. The potential impact on telephony of the ADSL separation filter is increased echo and sidetone. BT Laboratories has pioneered the design of filters that mitigate this problem [9,10].

Three transmission methods were originally proposed for use for ADSL and are now all being considered for VDSL. These options are briefly outlined in sections 4.2 — 4.4 below.

4.2 Quadrature amplitude modulation (QAM)

QAM is a well-established and documented modulation technique used extensively in voiceband modems and microwave radio systems. The source data is split into two half-rate streams which are then modulated on to a pair of orthogonal carriers for transmission. The orthogonality is provided by a sine and cosine mixing function (modulator) which can be implemented digitally. The transmitter comprises a scrambler, bit-to-symbol encoder, transmit low-pass filter, modulator and digital-to-analogue converter (Fig 13).

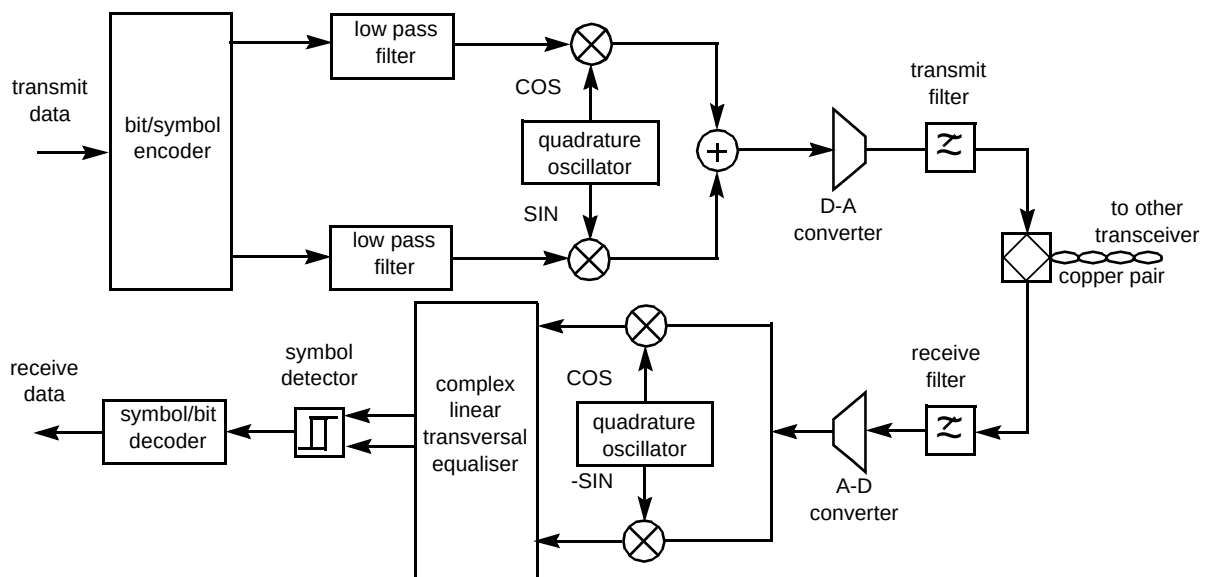


Fig 13 QAM transceiver.

At the receiver, the orthogonality allows the two-bit streams to be separated by demodulation for subsequent detection of data. Detection involves identifying the received 2-D complex symbol and then mapping this back into binary and descrambling the bit stream. The receiver also includes equalisation to compensate for the dispersion caused by the channel. In order to avoid aliasing in the analogue-to-digital conversion process, the front end of the receiver would typically operate at either 3 or 4 samples per symbol which implies the need for a fractional-tap linear equaliser that operates at 3 to 4 times the symbol rate. Linear equalisers are well known for causing noise amplification in data transmission systems. Hence, most early proponents of QAM ADSL examined the use of decision feedback equalisation (DFE).

4.3 Carrierless amplitude/phase modulation (CAP)

CAP is a 2-D transmission scheme similar to QAM. CAP has the same type of spectral shaping as QAM and can be made compatible with QAM. However, instead of orthogonal carriers being generated by a sine and cosine mixer, modulation of the two half-rate bit streams is achieved via two digital transversal bandpass filters that have impulse responses which are a Hilbert pair (same amplitude characteristics but phase response differs by $(\pi/2)$). The transmit signal is then formed by combining the two outputs of the two digital filters (Fig 14).

The receiver must perform an equalisation function and can use the same type of equaliser structures used for QAM, e.g. two parallel linear adaptive filters with fractionally spaced tap coefficients and possibly a complex adaptive feedback filter (DFE). The front-end fractional tap filters in such a receiver also provide the matched filtering function to separate the incoming line signal into the two separate orthogonal components. The feedback filter section of a

DFE can result in error propagation, but techniques such as Tomlinson precoding can alleviate this problem and have been studied in the context of CAP transceivers [11].

4.4 Discrete multi-tone modulation (DMT)

DMT is a form of multicarrier modulation. DMT divides time into regular 'symbol periods', each of which will carry a fixed number of bits. The bits are assigned in groups to signalling tones of different frequencies. Hence, in the frequency domain, DMT divides the channel into a large number of sub-channels. The line capacity varies with frequency and those sub-channels with higher capacity are assigned more bits. The bits for each tone or sub-channel are converted into a complex number which will set the tone's amplitude and phase for the symbol period. Hence, conceptually, DMT can be thought of as a bank of contiguous QAM systems operating simultaneously in parallel, each with a carrier frequency corresponding to a DMT sub-channel tone frequency (Fig 15). Thus the DMT transmitter essentially modulates data by forming tone bursts for a number of frequencies, adding them together and sending them to line as a 'DMT symbol'.

The multicarrier modulation (and demodulation) requires orthogonality between the various sub-channels. This can be implemented in an efficient all-digital realisation by exploiting fast Fourier transform (FFT) methods [12] (Fig 16). Each narrow sub-channel has the advantage that the cable characteristic is approximately linear across the entire sub-channel, and thus pulse dispersion (and the need for equalisation in the receiver) within each sub-channel is minimal. The energy from an impulsive noise event will modify a received symbol but the FFT 'spreads' the effect over the large number of sub-channels within the FFT window. The event is therefore less likely to cause an error.

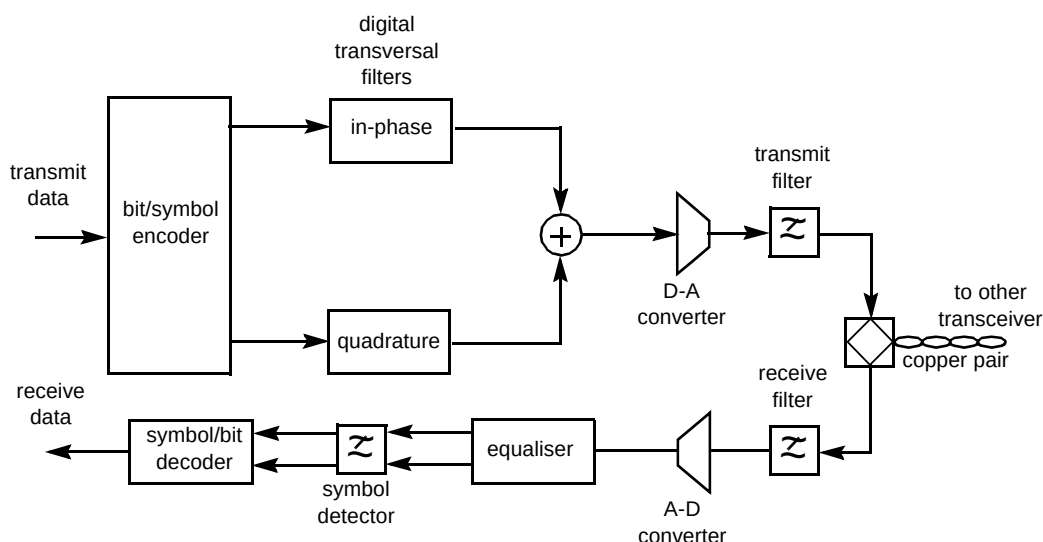


Fig 14 CAP transceiver.

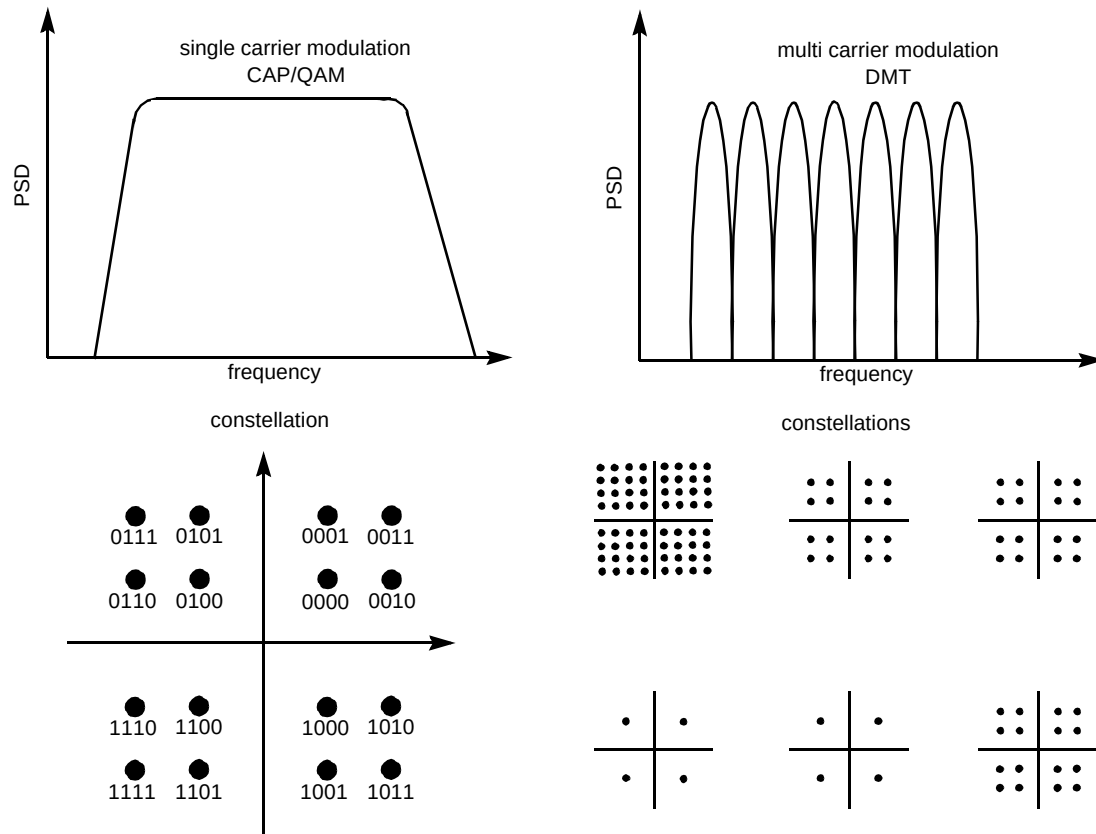


Fig 15 Single carrier and multi-carrier modulation.

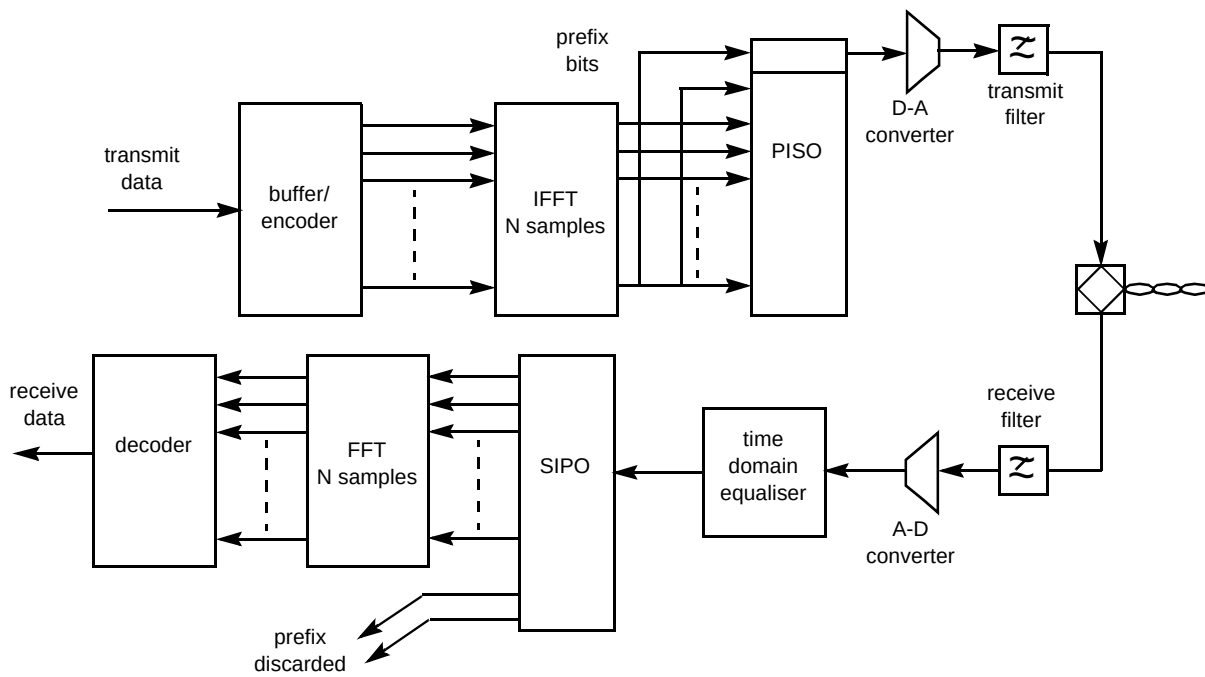


Fig 16 DMT transceiver.

Another form of multicarrier modulation, being explored by some manufacturers, uses a wavelet transform instead of the FFT. This is known as discrete wavelet multi-tone (DWT). With both DMT and DWT the number of bits of data sent over each sub-channel can be varied adaptively depending on the signal and noise levels in each sub-channel. Not only does this allow performance to be maximised for each particular loop but it can be a particular advantage for loops which suffer crosstalk or radio frequency interference, since the system can adapt automatically to avoid regions of frequency spectrum that are unsuitable for data transmission (Fig 17).

4.5 Improving performance in the real network

Although ADSL systems use a higher absolute transmit power (at + 20 dBm) than previous copper pair digital transmission systems, this power is spread over a wider bandwidth. Consequently, the power spectral density (PSD) can be specified to be at levels similar to existing services. The result is that crosstalk between, say, ADSL and ISDN will be no different from self-crosstalk between two ISDN systems running in the same cable. In fact, the ability of DMT ADSL to allocate transmit power in a non-uniform manner across the sub-channels (within the constraints of upper limit mask) can give even greater spectral compatibility. The system will simply use less power at those frequencies occupied by other services. CAP-based ADSL can use the design of the transmit filter to provide a degree of non-uniform transmit PSD.

Some manufacturers are implementing a power back-off mechanism whereby on shorter loops, where required performance can be achieved easily, the transceiver will

reduce the overall transmit power level to as little as - 10 dBm.

ADSL and VDSL transceivers employ many of the state-of-the-art modem techniques that were previously only possible at voiceband signal processing speeds. As such, ADSL and VDSL are far more sophisticated and adept at coping with the copper pair transmission environment than currently deployed systems such as ISDN or pair gain systems. For example, some countries (not the UK) have an embedded copper network where some pairs have been jointed by twisting pairs together rather than using more robust connection techniques. Microinterruptions are a particular concern for such countries. It is known that the short break in transmission caused by such micro-interruptions can cause ISDN transceivers to reset, effectively causing the call to be dropped. ADSL transceivers exist that will flywheel through an interruption of up to 5 seconds (far longer than 'micro') without resetting to a cold start-up. Basically the system is designed to freeze its adaptive elements (such as equalisation and timing acquisition circuits) during the break, and then unfreeze them as soon as the break reconnects. Transmission errors can occur during the flywheel period, but for many 'micro' breaks these errors will be corrected by a forward error correction (FEC) code.

DMT is a modulation technique that has a degree of inherent immunity to impulse noise. Impulses occurring within a DMT symbol period will often not cause an error. The use of FEC and interleaving gives DMT transceivers resilience against even longer impulsive noise bursts. In designing the parameters of the FEC and interleaving, there

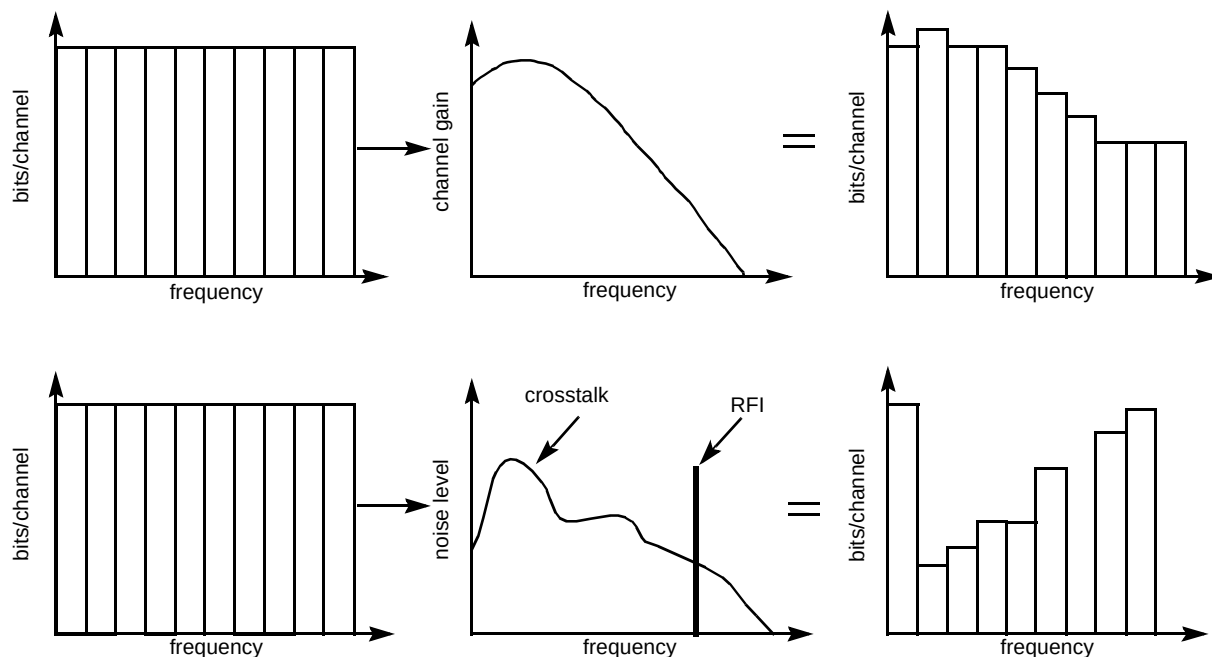


Fig 17 DMT bit allocation.

is a trade-off between degree of protection and the resulting transmission latency. The current ANSI standard for DMT ADSL transceivers will protect up to 3 DMT symbols. This will enable it to survive an impulsive noise hit of up to 500 μ s without causing an error. The latency is around 20 ms on the downstream channel. The interleaving may be bypassed in standard-compliant transceivers for applications where the latency of transmission is more important than bit integrity. It should be noted that some services that will be carried over ADSL may have a degree of self-protection against transmission errors. For example, with a video on demand service, the video compression scheme may include its own error concealment features.

The main purpose of the FEC and interleaving is to provide error protection against impulsive noise. Laboratory measurements of ADSL and VDSL performance against crosstalk will include the performance enhancement due to the FEC and interleaving (typically around 3 dB). However, in designing deployment guidelines for ADSL it is prudent to determine the systems range in a pessimistic crosstalk environment with the FEC gain 'factored out'. This will ensure that when the system is deployed on the longest of lines, it will still have the power of the FEC in reserve to deal with impulse noise events, many of which could be sourced from within the customer's premises. If this philosophy is adopted, then the system deployment range can only be increased by enhancing its performance against the steady-state background noise such as crosstalk. One way of achieving this is to use trellis coding. This will give an additional 2.5 dB performance gain when used in conjunction with the FEC. This is an option for ADSL standard-compliant systems, with some manufacturers choosing to omit this additional complexity in early implementations. A further way of increasing range is to use echo cancellation as discussed in section 5.1.

The front-end of a QAM or CAP transceiver usually contains a fractional-tap equaliser. This oversampling digital filter uses an adaptation algorithm to minimise noise at the decision device in the receiver. In addition to minimising the intersymbol interference (i.e. performing equalisation), this filter can also synthesise a notch in its passband to reduce the impact of narrowband noise such as RFI. A DMT system has even greater flexibility to deal with multiple RFI tones by simply not assigning information to those sub-channels where RFI exists. RFI is more likely to be picked up on lines in rural areas where more overhead cabling exists. The steady state adaptation of equalisers and DMT bit-swapping algorithms provide a degree of tracking of time-varying RFI noise. With a DMT-based VDSL system it is not necessary to have the transceiver 'react' to RFI by adapting to it during operation. For example, transmission in the DMT sub-channels that correspond to amateur radio bands can be prohibited permanently by taking this into account in the design of the transceiver [7]. This principle of blanking out transmission at certain

frequencies has had to be used by some cable companies to avoid interference (to and from) amateur radio enthusiasts.

4.6 CAP/QAM versus DMT

The transmission options described above each have advantages and disadvantages. These have been the subject of considerable debate in the ANSI and ETSI standards fora, both for ADSL and VDSL. It has been recognised that QAM and CAP systems can be implemented in VLSI with similar complexity and achieve similar performance [13], hence the real debate has been between single-carrier (CAP and QAM) versus multi-carrier (DMT) technology. Some of the arguments and counter arguments that have been cited for each technology are listed below:

- DMT's ability to adaptively shape the allocation of information and transmit power across a given bandwidth enables it to better approach optimum performance than is theoretically possible with CAP/QAM,
- the complex start-up procedure required for DMT could result in longer activation times than is possible with CAP/QAM,
- DMT offers greatest performance advantages over CAP/QAM at high transmission rates — the shorter loops at higher rates have more usable bandwidth giving DMT a greater degree of freedom (more usable sub-channels) for flexible adaptation,
- DMT uses block (FFT) processing that results in transmission delay (latency) — this may violate some telco/international specifications for services such as telephony and ISDN at ADSL speeds, whereas it is not an issue at VDSL speeds,
- DMT is more adept than CAP/QAM at coping with multiple RF interferers,
- a high peak-to-average ratio in the DMT transmit signal can lead to clipping noise and expensive analogue to digital conversion — part of the power of the error correction may be used up in dealing with this additional noise,
- it is simple for DMT to meet an arbitrary transmit power mask specification to meet spectral compatibility requirements,
- echo cancellation in DMT systems is non-trivial,
- DMT has a greater inherent immunity to impulsive noise than CAP/QAM,
- CAP/QAM can use simpler forward error correction than DMT,
- DMT requires minimal equalisation with slower signal processing rates than CAP/QAM,

- DMT hardware can be more easily programmed to support a variety of upstream and downstream data rates with potential for on-line reconfiguration,
- DMT need be no more complex to implement than CAP (in terms of signal processing MIPS per area of VLSI real estate), although it may be more complicated to understand and design, in particular the start-up sequence,
- DMT has the potential for more sophisticated embedded line monitoring functionality (e.g. for cable sheath management).

It is left to the proponents of each technology to debate the validity, importance and cost implications of the above statements. In the early days of ADSL development, CAP/QAM prototypes performed worse than DMT, but some of the core 'bit-pump' signal processing was more highly integrated. Manufacturers' development programmes for subsequent generations of each technology are providing the incremental steps towards the telcos' 'ADSL target' of high performance and low cost. The picture will change rapidly over the next two years or so as more manufacturers enter the ADSL/VDSL market. Increasingly there will be a focus on differentiators such as quality of implementation, added functionality and network management capability.

5. Achievable transmission performance

5.1 ADSL performance

In the presence of a very pessimistic noise source (HDSL and ISDN crosstalk plus RFI), DMT ADSL is anticipated to be able to operate (with a 6 dB performance 'safety' margin) over customer lines having an insertion loss of up to about 51 dB at 300 kHz [14]. This is the consensus of a variety of telcos and manufacturers all performing independent performance analysis for ADSL standardisation work [15, 16]. This corresponds to a range of around 3.6 km to 5.75 km depending on cable gauge (see ANSI loopset [6, 17]) for 2 Mbit/s downstream transmission with a 16-kbit/s control channel. This level of performance puts in excess of 90% of BT customers within range of 2 Mbit/s ADSL. At this lower end of the range of ADSL data rates, performance of CAP systems has recently become comparable. The performance of this 'basic' system can be increased by the use of additional signal processing and transmission techniques. For example, the combined use of trellis coding, echo cancellation and optimised transmit power distribution [15] will increase the performance by around 5 to 6 dB, thus giving even greater reach and network penetration for an equivalent level of performance.

The performance of the base 2 Mbit/s ADSL system reduces by around 2 dB (about 130 to 220 m) if capacity for ISDN is also included in the control-channel payload of an FDM implementation. It should be noted, however, that ADSL using DMT modulation and/or interleaved FEC will

violate any 1.25 ms ISDN latency requirement. If the 2-Mbit/s payload of the ADSL system is to be transported in ATM cell format, then the ATM overhead (without significant OAM cell overhead) will reduce performance by around 0.8 to 1.8 dB (about 60 to 160 m).

In an equivalent pessimistic noise environment, systems operating with a 4-Mbit/s payload will operate over loops having an insertion loss of up to about 43 dB at 300 kHz [18]. This corresponds to a range of about 3.1 km on 0.4 mm cable or 4 km on 0.5 mm cable. Similarly, systems operating with a 6-Mbit/s payload will operate over loops having an insertion loss of up to about 38 dB at 300 kHz. This corresponds to a range of about 2.7 km on 0.4 mm cable or 3.6 km on 0.5 mm cable. Figure 18 and Table 1 summarise the results for basic DMT ADSL in the presence of pessimistic crosstalk noise. As with the 2-Mbit/s system, range, and hence network penetration, can be further increased for the 4- and 6-Mbit/s systems by the use of trellis coding, echo cancellation and optimised transmit power distribution.

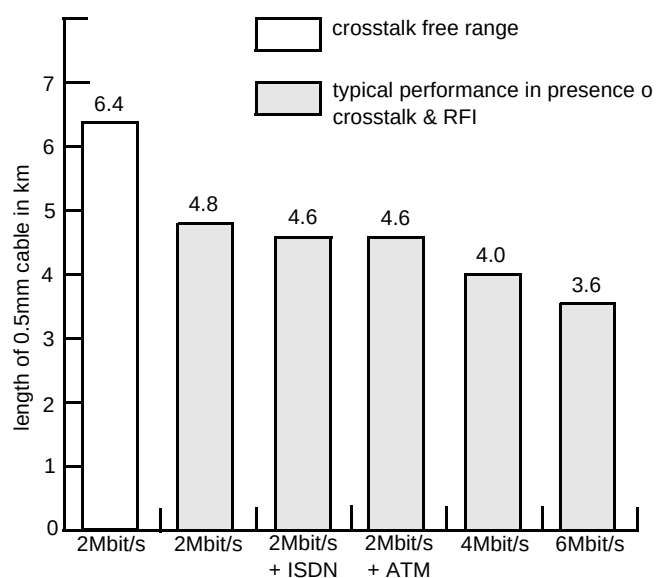


Fig 18 'Basic' ADSL performance in terms of range.

Table 1 'Basic' ADSL performance in terms of insertion loss at 300 kHz.

Insertion loss 0.5 mm Cu	No ISDN No ATM	With ISDN No ATM	No ISDN With ATM
2.048 Mbit/s payload	51 dB	49 dB	50 dB
4.096 Mbit/s payload	43 dB	42 dB	42 dB
6.144 Mbit/s payload	38 dB	36 dB	36 dB

It should be noted that the above performance measures are in the presence of pessimistic crosstalk and RFI. In the absence of this noise 2-Mbit/s systems could perform over loops having an insertion loss of around 67 dB to 70 dB at 300 kHz. This maximum range corresponds to over 99% of

BT loops. In practice, many customer lines will exist in a noise environment that lies somewhere between the pessimistic crosstalk and maximum range, scenarios. However, many telcos formulate planning and deployment rules on the basis of a worst-case scenario in order to give a more certain lower-bound on performance.

5.2 VDSL Performance

In contrast to ADSL, the number of current developments of actual VDSL transceivers is more limited (see Chow and Cioffi [7]). Hence, most performance estimates are based on simulation with only a few early VDSL prototype measurements having been published [19]. There is a good history of correlation between simulation results and actual implementation of copper network digital transmission systems spanning ISDN, HDSL and ADSL technologies. Therefore reasonable faith can be placed in such predicted performance. Channel models have been developed that have been proved by laboratory cable measurements to be accurate at VDSL operating frequencies (see section 3.2).

A reasonably pessimistic noise environment for VDSL is to have 12 FEXT interferers. Under these conditions, a 'basic' DMT system operating over 0.5 mm cable with EMC-legal transmit power level, FDM downstream/upstream channel separation and 8.8 MHz operating bandwidth shows the potential to achieve up to 25-Mbit/s downstream transmission over 1 km with a 6-dB performance margin [8]. This sort of performance may also be achievable when transmission in the amateur radio band frequencies is prohibited [7].

Field measurements of actual installed network cables have shown, unsurprisingly, that there is variance in cable characteristics resulting in some actual cables being better and some worse when compared to the nominal laboratory cable or model. The cumulative distribution of customer line lengths flattens out at high penetration rates [20]. Hence, a large percentage reduction in system deployment range, to accommodate worst-case cables, results in a much smaller reduction in percentage customer penetration. To enable many out-of-range customers to receive a lower rate service, it would be possible to have the system rate adapt in the way that voiceband modems do. A few coarsely quantised options (e.g. half and quarter rate) would be simpler than a continuously variable selection. As the VDSL transmitter moves closer to the customer, such as in a fibre-to-the-kerb architecture, the same basic DMT system could potentially deliver more than 40 Mbit/s over distances up to 300 m [8].

As with ADSL, the performance of this basic DMT VDSL system can be increased by around 6 dB via the use of trellis coding, echo cancellation, optimised transmit

power distribution and larger DMT sub-channel constellations. Although greater transmit power would further increase performance, the scope for this is limited with VDSL, since transmit power significantly greater than + 10 dBm would be prohibited by EMC emission constraints and worst-case pair balance. This is lower than ADSL transmit power levels because VDSL operates at frequencies where the copper cable becomes a more efficient radiator. However, it may be possible to use a tapered spectrum with higher spectral densities at lower frequencies where the worst-case pair balance is greater. It is technically feasible to use more signal processing power to extend the range of a VDSL transceiver beyond the distances quoted above. However, the size and power constraints of externally located VDSL equipment together with the inevitable cost constraints could prohibit the use of more sophisticated systems until future generations of highly integrated transceiver chips become available.

6. Status of the technology

6.1 Trials and equipment

Following measurements on the first ADSL prototypes in early 1993, ADSL has now left the laboratory and has been used in field trials of video on demand by telcos such as BT and Bell Atlantic. Around 30 other telcos are planning to trial ADSL during the coming year in Europe, the USA, South East Asia and Australasia [21]. These trials are planned to evolve from small-scale technology trials to much larger market trials. ADSL proved itself technically viable in a 60-customer BT video on demand trial in 1994 [22]. The transmission technology was very reliable and BT are now using ADSL to deploy a wide range of interactive services to 2000 customers in a market trial [23].

As manufacturers progress through the various stages of product development ADSL systems will migrate from stand-alone modems through to embedded line cards with comprehensive network management functionality. ATM and customer interface issues will be addressed. Some manufacturers will offer additional performance enhancing features such as trellis coding and echo cancellation. In 1996 it is anticipated that there will be a multiplicity of ADSL vendors from which to choose [21, 24–30] with some of them offering compact VLSI transceiver implementations. VDSL systems currently only exist as initial prototypes but look set to follow a similar development trajectory to ADSL.

6.2 International standards

The details of the transmission technology for ADSL have been studied by ANSI since 1991. In January 1993 the

'ADSL Olympics' were held. This involved BT, Bellcore and Nynex performing measurements on early DMT, CAP and QAM prototypes. In Florida in March 1993, after considerable debate, ANSI working group T1E1.4 agreed to base its interface definition [6] on the DMT modulation method [31]. Work then progressed in this forum on the definition of the details of electrical characteristics, bit rate options, framing structures, on-line adaptation and reconfiguration, start-up and signalling requirements, impairments and testing, and maintenance aspects. The first issue of the ANSI ADSL standard was ratified early in 1995.

The ANSI ADSL standard [6] contains an annex which gives detail of issues which are specific to 2.048 Mbit/s ADSL requirements, of particular interest to Europe and Australia. The European Standards working group ETSI STC TM3 RG12 were highly instrumental in providing ANSI with information and text for this annex [17] which was edited by BT. This will assist European network operators who wish to acquire ADSL equipment based on standard silicon devices. ETSI have recently started their own work item with the intention of producing a European Technical Report (ETR) on ADSL.

The ADSL Forum is an international group of telcos and ADSL suppliers. It seeks to advance the rapid world-wide deployment of ADSL by facilitating development of interoperable, end-to-end, ADSL-based network components. Other groups such as the Digital Audio-Visual Council (DAVIC) are examining the overall network architecture for interactive broadband services. ANSI and ETSI are producing standards and reports covering ADSL chip requirements. The ADSL Forum will bridge the gap between silicon chips and networks by examining the interface requirements for ADSL 'boxes' and line-cards. Much of its work will involve interpreting the relevant work of other fora and standards bodies and forming voluntary technical guidelines (particularly on interfaces) for ADSL systems. BT is represented on the board of directors of the ADSL Forum.

Both ETSI and ANSI have begun technical studies on VDSL and a standard has been proposed [32]. BT is leading the work package on VDSL within ETSI and is highly active within ANSI. The ADSL Forum and DAVIC have also discussed aspects of VDSL.

7. The future for ADSL and VDSL

ADSL has left the laboratory and has proved its ability to perform in a real network environment as part of video on demand trials. The next stages in the evolution of

this technology will see manufacturers working to improve performance, functionality and levels of integration as well as reducing costs. VDSL will also follow a similar trajectory from today's early prototype developments towards fully engineered and integrated products. International standards bodies will help stimulate and accelerate such developments. For their part, telcos will be working to develop slick operational processes for the planning, deployment and management of these advanced access technologies.

For years, the 'bread and butter' business of telcos has been delivery of POTS using an access network that is largely copper based. More recently processes have evolved to accommodate digital transmission over copper for private circuits, ISDN and pair-gain applications. ADSL represents another step in this evolution, and operational processes, workforce skills and management systems may be gracefully migrated to accommodate the use of ADSL [33]. Broadband hybrid fibre/copper networks based on optical fibre and VDSL technologies may borrow from the experience gained from the development of telephony over a passive optical network (TPON) [34].

There are numerous access network strategies [35, 36] and applications [37, 38] that are well served by ADSL and VDSL. Various telcos will choose to employ these technologies in different ways depending on their local competitive and regulatory environment.

ADSL provides a per-customer connection in which for the vast majority of customers the network infrastructure (copper pairs) is already in place. Capital expenditure to create an ADSL line is matched by a potential new revenue stream. Thus ADSL reduces up-front speculative expenditure by deferring equipment deployment until the customer requests service, i.e. 'just in time' (JIT) provision of service with reduced sensitivity to service take-up. This makes ADSL ideal for trials of new services.

Initially, the major trials and applications using ADSL have focused on the residential market with services such as video on demand. However, there is now an increasing trend to consider ADSL for business applications. The ability of standard-compliant ADSL chips to offer bi-directional data rates in excess of 500 kbit/s, together with POTS and the high-speed downstream channel, could be attractive for offering data services such as high-speed Internet access, SMDS and frame relay to businesses and teleworkers. The downstream channel will also be used for remote database access, e.g. for estate agents, sales people, etc). ADSL is useful for offering services having limited appeal within a confined geographical area since it serves customers as opposed to areas [39].

New multimedia interactive services may be rolled out and marketed on a town-by-town basis. Customers requesting these services can be provisioned rapidly with ADSL. Then, when a reasonable cluster of customers in an area have taken up service [39], a shared infrastructure architecture such as fibre with VDSL copper tails may be attractive and the telco can be proactive in taking fibre to those customers. The ADSL and VDSL delivery systems could provide the same customer interface. Hence the customer's internal wiring and set-top box remain unchanged as the access network evolves to include fibre.

In relative terms, the capacity of the copper network is limited. However, ADSL and VDSL will facilitate the process of taking new services and optical fibre technology increasingly closer to the home, the first step towards national superhighways.

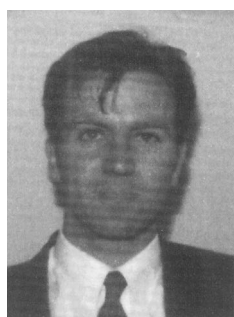
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References

- 1 Cook J W: 'Measurement and simulation of transmission line primary line constants', BT Internal Memorandum (June 1994).
- 2 Foster K T: 'On the capacity of the access network and the potential for VDSL — multi-megabit/s transmission in BT's metallic access network', BT MSc in Telecom Engineering Dissertation (March 1995).
- 3 Waring D L, Lechleider J W and Hsing T R: 'Digital subscriber line technology facilitates a graceful transition from copper to fiber', IEEE Communications Magazine (March 1991).
- 4 Chow P S, Tu J C and Cioffi J M: 'Preliminary feasibility study of a multi-carrier transmission system for the proposed ADSL data service', ANSI T1E1.4 contribution, T1E1.4/90-21 (December 1990).
- 5 Cole N G et al: 'A low-complexity, high speed digital subscriber loop transceiver', BT Technol J, 10, No 1, pp 72—79 (January 1992).
- 6 American National Standards Institute (ANSI) Working Group T1E1.4, Document No. T1E1.4/94-007R8: 'Asymmetric digital subscriber line (ADSL) working draft Standard', (will be published as the T1.413 Standard during 1995).
- 7 Chow P and Cioffi J: 'Broadband digital subscriber lines', IIR Maximizing Copper Conference, London (March 1995).
- 8 Olshansky R and Veeneman D: 'Broadband ADSL', ADSL Forum (Contribution ADSL Forum '95-016) (March 1995).
- 9 Cook J W: 'Telephony Transmission and Splitters, Passive and Active', ANSI T1E1.4/94-043 (February 1994).
- 10 Cook J W and Sheppard P J: 'ADSL and VDSL splitter design and telephony performance', IEEE JSAC (to be published December 1995).
- 11 Zervos N A: 'High-speed carrierless passband transmission over the local cable network', 703.7.1 Globecom, San Diego, USA (December 1990).
- 12 Bingham J A C: 'Multicarrier modulation for data transmission — an idea whose time has come', IEEE Communications Magazine, pp 5—14 (May 1990).
- 13 Samueli H: 'VLSI complexity comparison of a WAM versus CAP ADSL transceiver', ANSI T1E1.4/92-4 (August 1992).
- 14 Kirkby R: 'Loop-set for 2.048 Mbit/s ADSL with noise model A', ETSI STC TM3 WG1 (RG12) TD No 4, Ipswich, UK (June 1994).
- 15 Etter P J: 'Modelling results for DMT ADSL transmission', ETSI STC TM3 WG1 (RG12) TD No 12, Ipswich, UK (June 1994).
- 16 Genossar M J: 'Simulation results for 2048 kbit/s ADSL with noise model A', ETSI STC TM3 WG1 (RG12) TD No 10, Ipswich, UK (June 1994).
- 17 Foster K T: 'Text for annex C', ANSI T1E1.4/94-091R2 (June 1994).
- 18 Kirkby R H: 'Results of performance simulations of various ADSL configurations', IEE Colloquium on High Speed Access Technology and Services, including Video on Demand, London, UK (October 1994).
- 19 Harman D D, Huang G, Im G H, Nguyen M H, Werner J J and Wong M K: 'Local distribution for interactive multimedia TV to the home', IEEE, pp 175—182 (1994).
- 20 Du Chaffaut G and Autrique F: 'France Telecom network features', IIR Maximizing Copper Conference, London (March 1995).
- 21 Faw R: 'Global ADSL update', Keynote Address IIR Maximizing Copper Conference, London (March 1995).
- 22 BT News Release NR9479: 'BT to trial interactive TV', (November 1994).
- 23 Interview with BT's Rupert Gavin by Ben Keen: 'BT's VoD: Tip of the ICEberg', ATM Magazine, p 10 (February 1995).
- 24 Dahl K A: IIR Maximizing Copper Conference, Vendor Panel Session, London (March 1995).
- 25 Whelan G: IIR Maximizing Copper Conference, Vendor Panel Session, London (March 1995).
- 26 Sallee D: IIR Maximizing Copper Conference, Vendor Panel Session, London (March 1995).
- 27 Tatler S: IIR Maximizing Copper Conference, Vendor Panel Session, London (March 1995).
- 28 Rowe J: IIR Maximizing Copper Conference, Vendor Panel Session, London (March 1995).
- 29 Moon D Y: IIR Maximizing Copper Conference, Vendor Panel Session, London (March 1995).
- 30 Astrand S: 'Examining the interdependence and integration required between switching transmission and the access network in order to cost effectively deliver broadband services', IIR Maximizing Copper Conference, London (March 1995).

- 31 Cioffi J M et al: 'A DMT proposal for ADSL transceiver interfaces', Amati Communications, Palo Alto, ANSI T1E1.4/92-174 (August 1992).
- 32 Cioffi J M and Bingham J A C: 'Workplan for broadband digital subscriber lines project evolution', ANSI T1E1.4/95-011 (February 1995).
- 33 Adams P F, Foster K T and Hind M: 'The operations and maintenance aspects of high-speed copper access transmission systems', ICC94, New Orleans, USA (1994).
- 34 W K Ritchie (Ed): 'Local Telecommunications Networks', BT Telecommunications Series, Chapman and Hall (1993).
- 35 Starr T: 'The strategic relationship of HFC and ADSL', The ADSL Forum (December 1994).
- 36 Young G: 'ADSL from concept to BT trial and beyond', IIR Maximizing Copper Conference, London (March 1995).
- 37 Maxwell K: 'Canonical applications', ADSL Forum 95-022 (March 1995).
- 38 Handzel M: 'ADSL services', ADSL Forum 95-028 (March 1995).
- 39 Interview with Ray Smith of Bell Atlantic by D Kline: 'Align and Conquer', Wired, p 110 (February 1995).



Gavin Young received the BSc degree in electrical and electronic engineering from the University College of Swansea, Wales, in 1985, and the MSc degree in Communications Engineering from the Imperial College of Science and Technology, London University in 1986. In 1986 he joined BT Laboratories working initially on the design of an ISDN basic access transceiver chip. He was subsequently involved in research and development for high-rate digital subscriber loop (HDSL) transceivers with emphasis on the design and simulation of digital transmission techniques and adaptive DSP architectures. In 1992 he was involved in leading feasibility work on the use of asymmetric digital subscriber loop (ADSL) technology for a video-on-demand service. His current roles are copper access technical area leader and project manager for development work on ADSL. He is involved in progressing work on ADSL and its higher speed variants within ANSI and also the ADSL Forum of which he is a director.

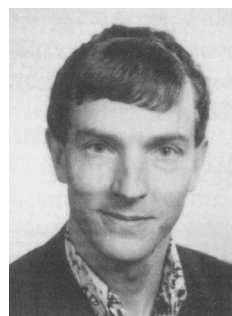


Kevin T Foster graduated in 1982 with an honours degree in electronic and electrical engineering and diploma of industrial studies from Loughborough University. He received his Masters Degree (with distinction) in telecommunications engineering from the University of London in 1995.

He joined BT Laboratories in 1984 and worked on digital line interfaces for the Monarch PABX. Subsequently he was involved with the collaborative design of an ISDN 'U' interface transceiver chip with Mitel. During the late 1980s he was closely involved with the design, specification and performance verification of BT's prototype HDSL transceivers. He has represented BT at numerous international standards meetings in the US and Europe.

He is currently co-ordinating standards work in Europe on the feasibility of broadband metallic access systems and is BT's representative in ETSI TM3 WG1. At BT Laboratories his current role is project technical leader for a team of engineers engaged in broadband transmission over metallic access wire-pairs.

He is a member of the IEEE and an associate member of the IEE.



John Cook joined BT as a student in 1974, and graduated in electronics from Queen Mary College, University of London in 1978. He then joined the BT Laboratories and until 1984 worked on various aspects of speech codecs and POTS analogue line interface design. In 1987 he joined a team developing an ISDN basic rate transceiver chip.

Since 1988 he has been an adviser on copper access systems with particular emphasis on signal processing and analysis. During this time he has made significant contributions to the study of impulsive noise and its effects on digital access transmission systems and contributed to standards work on HDSL, CDDI, ADSL and VADSL both in Europe and the USA.

